Advantage Rules at Matched Accuracy: Noise and the Accuracy–Diversity Frontier in RL with Verifiable Rewards

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Abstract

Comparisons of reinforcement learning with verifiable rewards (RLVR) often mix learning speed with repeated-sampling success. We compare pass@16 at matched pass@1 using interpolated references and registered tests with bootstrap intervals over problems. Across the tested correctness-only rules, budgets, and Qwen model sizes, the evidence supports changes in the rate of progress along a shared pass@16-versus-pass@1 curve within each setting, rather than changes in the curve. We derive expected per-problem updates and exact second moments under independent on-policy sampling. A 1.5B replication trained and evaluated at 1,024 tokens meets all four average equivalence tests. A 7B replication at 2,048 tokens meets three; the direct contrast between negative- and positive-sample reinforcement is not interpreted under the clamping guard. Fixed-budget rollout allocation meets no registered frontier-improvement criterion. A minimum-hint credit pilot gives -0.01 points, 95% $[-1.24, +0.89]$, and is dropped by its selection rule. The answer budget can create apparent frontier gains: re-evaluation reduces the supervised warm-start step-0 gain from $+4.77$ points, 90% $[+3.45, +6.07]$, at 512 tokens to $+2.15$, 90% $[+1.10, +3.20]$, at 1,024. Matched-accuracy conclusions remain conditional on answer budget, reference support, and observed training seeds.

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1. Introduction

RL with verifiable rewards trains a language model on automatically checked answers. For binary rewards, many methods modify the advantage in $g = \sum_i A_i \nabla \log \pi(y_i)$, changing the weights assigned to correct and incorrect samples. Positive- and negative-sample reinforcement are a useful test case: they use only correct or only incorrect samples, respectively, yet have equal expected policy-gradient updates under on-policy sampling (Zhu et al., 2025; Williams, 1992; Greensmith et al., 2004). Normalization changes (Liu et al., 2025) and pass@k objectives (Walder & Karkhanis, 2025; Thrampoulidis et al., 2025) provide further comparisons. We study whether these choices change the accuracy–diversity trade-off measured by pass@1 and pass@k.

Fixed-step comparisons mix changes in single-answer accuracy with changes in repeated-sampling success. At matched pass@1, we compare a method's measured $\widehat{P}_k$ with a reference $R(a)$ interpolated at the same estimated accuracy $a = \widehat{P}_1$; the gain is $\widehat{P}_k - R(a)$. For method m, a trajectory is the ordered sequence of checkpoint measurements $\mathcal{T}_m = \{(\widehat{P}_1(\pi_{m,t}), \widehat{P}_k(\pi_{m,t}))\}_{t \in \mathcal{K}}$ at fixed decoding settings and answer budget. An envelope is the pointwise maximum of supported reference curves. Here $E_B(a) = \max\{L_{\text{steps}}(a), L_T(a)\}$ combines linear interpolation along Dr. GRPO's checkpoint trajectory with interpolation through a late-checkpoint temperature sweep; the maximum includes interpolants supported at a. Matched-accuracy comparisons have precedent in DARS (Yang et al., 2025). Our protocol adds pre-registered decision criteria and problem-level bootstrap intervals that recompute the reference under resampling.

We derive the mean and exact second moment of correctness-only updates under independent on-policy sampling and bounded reward-dependent stopping, and test whether changes in advantage rules or successful-sample availability alter pass@16 at matched pass@1. The registered comparisons support changes in progress rate along a shared curve within

each tested setting: all four contrasts replicate at 1,024 tokens on 1.5B, and three replicate at 2,048 tokens on 7B; the direct 7B NSR–PSR contrast is not interpreted under the clamping guard (Section 8). Fixed-budget allocation supplies more successes to pilot-zero problems but fails the registered improvement rules and its model forecast (Section 6). Answer-budget checks reduce the supervised warm-start step-0 gain from +4.77 points at 512 tokens, 90% [+3.45, +6.07], to +2.15 at 1,024 tokens, 90% [+1.10, +3.20], qualifying the apparent gain as largely a budget effect (Section 7). [Check 21: remaining human-solution seeds and the pooled budget-effect verdict.] Minimum-help process credit meets its mechanism gates but fails pilot selection (Section 10). [P22: results of external versus self prefix guidance.] Together, these comparisons distinguish faster accuracy progress from a supported change in the pass@16-versus-pass@1 relation.

At matched pass@1, pass@k is a concave average of per-problem success probabilities. For Dr. GRPO, a group of eight rollouts with no correct answer has zero advantages, so multiplicative reweighting cannot make that group contribute an update. This motivates replay and larger groups as interventions on the availability of successful samples. Coverage-weighted replay gives a mean gain of +1.19 pass@16 points, below the registered +2 threshold. Groups of 32 and a supervised warm start reach higher accuracy after 150 steps, but their mean endpoints lie beyond the available reference range; the longer-reference comparisons remain interim.

We also test a descriptive model of per-problem log-odds learning, with one transfer coefficient fitted on Dr. GRPO. Using base-model success counts and each checkpoint's observed pass@1, it predicts aggregate pass@16 with test-set RMSE of 1.21 points for nine further Qwen rules and 1.90 points on OLMo. It fails after the supervised warm start and on the allocation intervention it was used to design (Section 9), so we treat it as a conditional description rather than a mechanism.

The core registered comparisons and their outcomes are summarized in Appendix B. Interim estimates and exploratory diagnostics are identified in the text.

2. Setup and the matched-accuracy protocol

For a problem x with reward $r \in \{0, 1\}$, the policy's success probability is $p_x = \Pr_\pi(r = 1 \mid x)$, the score of an answer is $\ell(y) = \nabla_\vartheta \log \pi(y \mid x)$, and $F_r = \mathbb{E}[\|\ell\|^2 \mid r]$ is the mean squared score norm of correct ($r = 1$) and incorrect ($r = 0$) answers. Unless stated otherwise, a training step samples $G = 8$ rollouts for each of 16 problems. Dr. GRPO uses $A_i = r_i - \bar{r}$ within a group (Liu et al., 2025); positive-sample reinforcement (PSR) uses $A_i = r_i$ and negative-sample reinforcement (NSR) $A_i = r_i - 1$ (Zhu et al., 2025). Over N problems,

$$P_k = \tfrac{1}{N} \textstyle\sum_x \left[1 - (1 - p_x)^k\right],$$
$$\widehat{P}_k = \tfrac{1}{N} \textstyle\sum_x \left[1 - \binom{n_x - C_x}{k} \Big/ \binom{n_x}{k}\right],$$

where C_x counts correct answers among n_x independent samples and $\widehat{P}_k$ is unbiased for $1 \le k \le n_x$ (Chen et al., 2021). The main Qwen replay evaluations use $n_x = 32$.

Proposition 2.1 (pass@k at matched pass@1 measures spread). $P_1 = \mathbb{E}_x p_x$ and $P_k = \mathbb{E}_x f_k(p_x)$ with $f_k(p) = 1-(1-p)^k$ strictly concave for $k > 1$. Among distributions of $\{p_x\}$ with the same mean, a mean-preserving spread lowers P_k; P_k is largest when all p_x are equal.

Protocol. We identify the reference for each comparison. For the primary coverage-weighted and base-seeded replay contrasts, $E_B(a) = \max\{L_{\text{steps}}(a), L_T(a)\}$ is the larger of the linear interpolations in pass@1 along Dr. GRPO's trajectory (base model and steps 50, 100, 150) and its step-150 temperature sweep ($T \in \{1, 1.15, 1.3\}$), each within its supported range. NSR contrasts use PSR as the reference; uniform-replay and unlikeliness audits use Dr. GRPO's trajectory alone. Interim longer-training contrasts use the available step-250 trajectory. Gains are averaged over qualifying checkpoints within each run, then over seeds. Intervals use 2,000 bootstrap draws over problems, with the reference recomputed in every draw, conditional on the observed seeds. Observed checkpoints are restricted to the reference's pass@1 range; bootstrap endpoint clamping is disclosed in Appendix B. We call a difference average equivalence if its 90% interval lies within ±2 points and say a method moves the frontier if its gain is at least +2 points with the 95% interval above zero in every seed.

3. Theory

Theorem 3.1 (Two moments of correctness-only rules). Let a rule draw answers i.i.d. from $\pi(\cdot \mid x)$, stop at a time $N \le n_{\max}$ that depends only on the rewards observed so far, and assign advantages that depend only on the rewards. Then $g = \sum_{i \le N} A_i \ell(y_i)$ satisfies

$$\mathbb{E}g = c(p)\nabla p,$$
$$\mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p),$$

where c, D_1, D_0, X and $\mathbb{E}N$ depend only on the rule and on p.

The proof conditions on the reward sequence, given which N and the advantages are fixed and $\mathbb{E}[\ell \mid r = 1] = \nabla p/p$, $\mathbb{E}[\ell \mid r = 0] = -\nabla p/(1-p)$ (Appendix A). For fixed groups the mean part is known (Davis & Recht, 2025), and the equality of the PSR and NSR means is classical baseline invariance (Williams, 1992; Greensmith et al., 2004). The second-moment part, with $F_0 \neq F_1$ and reward-dependent stopping, is what lets us predict and manipulate noise.

Corollary 3.2. With advantage a for correct and b for incorrect answers in a group of n, $\mathbb{E}g = n(a-b)\nabla p$. PSR and NSR have the same mean $n\nabla p$, and NSR equals PSR plus the zero-mean term $-\sum_i \ell(y_i)$; when $F_0 > F_1$, NSR has the larger second moment on every problem with $p < F_0/(F_0 + F_1)$.

Lemma 3.3 (Rarely solved problems receive sparse updates). Under Dr. GRPO with a group of n, the update on problem x is zero unless the group contains both a correct and an incorrect answer. This event has probability $1 - p^n - (1-p)^n \approx np$ for small p. Multiplicative reweighting of Dr. GRPO's advantages cannot make an all-failure or all-success group contribute an update.

A common frontier does not follow from the moment theorem. The expected update on each problem is collinear with ∇p_x, but changes in the per-problem coefficients can rotate the aggregate update through shared parameters (Barakat et al., 2026). The sparse-update lemma establishes a property of Dr. GRPO and its multiplicative reweightings; it does not establish a necessary condition for every frontier improvement. It motivates testing replay and larger groups as ways to change which problems receive realized updates.

4. Advantage rules at matched accuracy

Models. Qwen2.5-Math-1.5B (Yang et al., 2024): MATH training problems, 150 steps, answers up to 512 tokens, evaluation on 511 held-out MATH test problems. OLMo-2-0425-1B-SFT (Team OLMo, 2024): 300 steps, 256 held-out problems, 1,024-token answers. Our default update uses an on-policy sequence-sum loss with AdamW at 10^{-6}; the sample-reuse replication and replay interventions modify this setup. Label checks are described in Appendix C.

Conditional score norms and update second moments. At initialization, F_0/F_1 is 2.14 and 2.54 on OLMo and 2.7–3.4 on Qwen. On Qwen it falls to about 1.4–1.6 at steps 50–150, and the registered persistence criterion is inconclusive. In an exploratory diagnostic with F_0 and F_1 re-measured and interpolated during training, predicted and measured second moments agree within a factor of 0.69–1.18 in every 50-step window. Raw NSR/PSR second-moment ratios are about 6 on Qwen and 19–22 on OLMo. On Qwen, rescaling by Adam's previous per-coordinate scales reduces the ratio to 1.04–1.32.

Table 1. Average matched-pass@1 differences in pass@16 (points; 90% bootstrap intervals over problems, conditional on the training seeds). Qwen rows use 128 problems and seeds 5–7; OLMo rows use 256 problems and two seeds. The OLMo pass@4 verdict is inconclusive. Checkpointwise equivalence is not established.

Comparison	Model, seeds	Difference [90%]
NSR vs PSR	OLMo, 2	−0.30 [−1.28, +0.77]
NSR vs PSR	Qwen, 3	−0.20 [−0.97, +0.55]
PSR + matched noise vs PSR	Qwen, 3	−0.37 [−1.13, +0.46]
NSR at 2× learning rate vs PSR	Qwen, 3	−0.03 [−0.88, +0.88]
pass@4 objective vs Dr. GRPO	OLMo, 2	+1.32 [+0.24, +2.26]

NSR and PSR agree at matched accuracy. The two satisfy average matched-accuracy equivalence on both model families (Table 1). Their progress rates differ less consistently: the registered OLMo speed prediction fails because both methods reach the midpoint at 52 steps in seed 1; in seed 2, NSR takes 88 steps and PSR 41. An earlier Qwen on-policy experiment found slower NSR progress and equal final pass@64. A Qwen replication using the implementation of Zhu et al. (2025), including sample reuse, clipping and token averaging, did not reproduce the reported advantage in this setting. Across constant baselines $A = r - b$, progress follows the ordering by predicted second moment (rank correlation 0.9–0.97), with matched-accuracy differences within about two points of PSR's trajectory.

Noise injection reproduces the pooled early lag. On Qwen, we add a zero-mean random-sign term to PSR, calibrated to match the same-policy NSR second moment. Over steps 25–49, the registered progress index is $\phi = -0.11$, with a 90% interval $[-0.22, -0.05]$, where NSR is 0 and PSR is 1. PSR with noise is slightly slower than NSR in the pooled comparison, and seed heterogeneity is flagged. This perturbation reproduces the pooled early accuracy lag without identifying scalar noise magnitude as its unique mechanism. Doubling NSR's learning rate gives $\phi = 1.16$ relative to PSR at its original rate; at the matched doubled rate, NSR still lags by 4.7 accuracy points, 95% $[2.7, 6.3]$, over two seeds. The Adam diagnostic measures the lagged, unnormalized projection onto a population-gradient reference: its NSR/PSR ratio is 0.73–0.87.

Temperature control. Higher-temperature evaluations of the late Qwen checkpoint have a mean matched-

pass@1 difference of −1.42 pass@16 points, with a 90% interval [−2.19, −0.68], relative to the training trajectory. Entropy on fixed base-model text changes from 1.081 to 1.064, while entropy on policy-generated samples falls three- to fourfold. These diagnostics distinguish the tested temperature sweep from the training trajectory. Entropy on policy samples mixes changes in conditional token distributions with changes in the sampled contexts.

Pass@k, unlikeliness reward, and entropy. The pass@4 objective (Walder & Karkhanis, 2025) gives matched-accuracy differences of +1.6 and +0.2 points on Qwen in two seeds. Its OLMo comparison is inconclusive under the registered rule (Table 1). Unlikeliness reward (He et al., 2025) gives +0.14 points, 95% [−0.32, +0.62], over three seeds. Unlikeliness reward and entropy control depend on trajectory likelihood or token entropy and are outside the correctness-only theorem. In a one-seed pilot, entropy bonuses gave no gain, while entropy floors and adaptive control prevented accuracy progress.

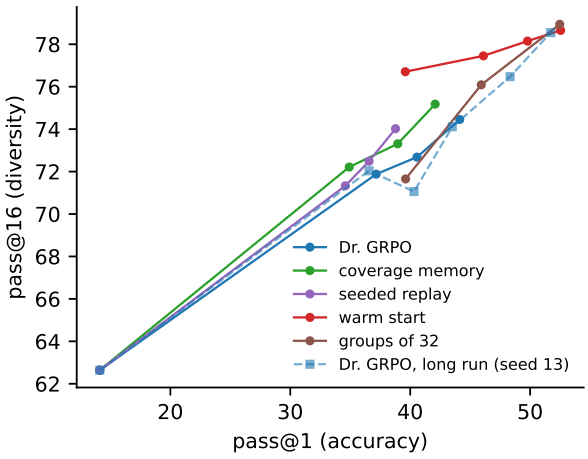


Figure 1. Seed-mean (pass@1, pass@16) trajectories on Qwen. G32 (two seeds) and the supervised warm start (three seeds) reach mean pass@1 52.5 at step 150, compared with 44.1 for Dr. GRPO. The interim longer reference discussed in the text reaches pass@1 51.7 at step 250 and does not cover these mean endpoints.

5. Replay, larger groups, and a supervised warm start

Dr. GRPO's zero advantages on all-failure groups motivate interventions that supply successful samples through replay or increased sampling. We evaluate replay, base-seeded replay, larger groups, and a supervised warm start on Qwen. The replay and warm-start arms use three seeds; the available larger-group results use two.

Replay. Coverage-weighted replay stores verified successes and adds a log-likelihood term, selecting problems in proportion to the posterior mean of $k\,p(1-p)^{k-1} = \partial f_k/\partial \log p$. Its mean gain is +1.19 points, 95% [+0.54, +1.79], against Dr. GRPO's trajectory-and-temperature envelope. Uniform replay gives +0.61, 95% [−0.04, +1.27], against the trajectory alone. The direct coverage-weighted versus uniform contrast is +0.66, below its registered +1 criterion. Coverage-weighted replay also lowers step-150 pass@1 by 2.05 points, despite its positive matched-accuracy gain.

The interim base-seeded replay analysis uses successes from 32 base-model samples per problem and gives +1.02 points, 95% [+0.23, +1.68], against the envelope, while lowering step-150 pass@1 by 5.35 points. The completed replay analyses meet the average-agreement criterion; the base-seeded estimate remains interim.

More rollouts, and a warm start. Dr. GRPO with groups of 32 (G32) uses more rollouts per step, while the supervised warm start on base-model successes (S32) adds a supervised stage before RL. Their higher accuracy at equal numbers of RL steps therefore comes with additional training resources. Endpoint comparisons are also limited by the accuracy range covered by the reference. For S32, Check 16 provides a supported comparison of the post-supervision initialization, step 0, with Dr. GRPO's trajectory at the same answer budget. The pooled matched-accuracy pass@16 gain is +4.77 points at 512 tokens, 90% [+3.45, +6.07], and +2.15 at 1,024 tokens, 90% [+1.10, +3.20]. The larger-budget gain is below half the original gain, meeting the registered criterion for reporting the warm-start advantage as largely a budget effect. A smaller positive initial gain remains. S32 checkpoints at steps 50–150 exceed the available Dr. GRPO pass@1 range at both budgets. These evaluations therefore establish a budget-dependent initial gain and leave the matched-accuracy position after RL unresolved (Section 7). Against an interim longer reference, Dr. GRPO seed 13 through step 250, G32 lies +0.60 points above it on 4 of 6 checkpoints, an interim point estimate without an interval.

6. Rollout allocation at a fixed budget

Pre-registration 18 tests whether reallocating sampled answers changes pass@16 at matched pass@1 without increasing the rollout budget. Each arm uses 128 rollouts per step on the same 16 problems as Dr. GRPO, denoted B. Training runs for 150 steps with 512-token answers in seeds 8, 9, and 10. Four pilot rollouts per

problem determine the success count c; the remaining 64 rollouts are distributed with a cap of 32 per problem. Pilot and additional samples are pooled before computing Dr. GRPO advantages.

Frontier-value allocation (AC) assigns additional rollouts in proportion to the registered model value

$$v(c) = \max\{0, \mathbb{E}[p^2(1-p)^2\{16(1-p)^{15} - \lambda\} \mid c]\},$$

with $p \mid c \sim \text{Beta}(c + 1/2, 4 - c + 1/2)$ and reference slope $\lambda = 0.37$. Only $c = 0$ has positive value under these settings; residual allocations after caps bind are distributed uniformly using largest-remainder rounding. Signal allocation (AV) instead weights additional rollouts by $\mathbb{E}[p(1-p) \mid c]$. Both retain the fixed normalizer. The HORA control, denoted HR, allocates greedily by posterior marginal hit gain under a Beta$(1, 1)$ prior and scales each problem's contribution by $8/N_x$, where N_x is its final group size.

Rule 1 compares AC with the trajectory-and-temperature envelope E_B; rules 2 and 3 compare it with the AV and HR trajectories. Their thresholds are +2 points for rule 1 and +1 for rules 2 and 3, with positive 95% intervals and Holm correction; rule 1 also requires positive gains in every seed. Gains are averaged over supported checkpoints and then equally over seeds; bootstrap intervals resample problems and condition on the trained seeds. Rules 1–3 are not met. AC gains +0.14 pass@16 points against E_B, 95% $[-0.36, +0.58]$; its 90% interval $[-0.28, +0.51]$ satisfies average agreement. AC versus AV gives +0.51, 95% $[+0.04, +0.99]$, below its effect threshold and nonsignificant after Holm correction. AC versus HR gives +0.00, 95% $[-0.52, +0.46]$. No confirmation run was triggered.

The descriptive mechanism diagnostics show that allocation operates as intended. Pilot-zero problems, defined by no success in the current pilot, receive a mean of 18.9 rollouts under AC. The reported fraction of their final groups containing a success is 50–54%, compared with 18–21% under AV. These diagnostics were reported without bootstrap intervals. Thus AC increases binary success signals on problems missed by the pilot without establishing a better matched-accuracy curve.

The registered point forecast check interpolates seed-mean curves at covered accuracy coordinates. Confirmation requires absolute error within 1.5 points of the forecast mean over those coordinates. HR and AV forecasts are confirmed under both model variants. AC has observed mean gain −0.05 points against forecasts of +4.12 from the frozen law and +2.46 from deeper initial counts; neither forecast is confirmed. The per-problem frontier model therefore fails on the allocation intervention it was designed to predict. More realized successes on rarely solved problems do not, in this experiment, produce the predicted change in the curve.

7. The answer budget as a confound

An answer budget b caps the number of generated response tokens. For M sampled answers, let $I_i(b)$ indicate whether answer i is stopped by this cap. We define the truncation rate as $\rho_b = M^{-1} \sum_{i=1}^{M} I_i(b)$. This rate matters even at matched pass@1: a length cap can affect different problems unequally, changing pass@16 at a fixed mean success probability. A comparison can therefore depend on response length as well as on the distribution of per-problem success probabilities.

At 512 tokens, Dr. GRPO has a reported truncation rate of 48%, whereas the human-solution arms of Preregistration 19 have rates of 6–7%. Both human-solution arms lie above its trajectory at this budget. Pre-registration 19 compares supervised training on human solutions for problems with observed base successes (HP) with a swap arm (HQ) that replaces some of those problems by problems with no observed base success. These strata refer to successes observed in 32 base samples per problem.

Check 16 re-evaluates existing checkpoints at the larger answer budget, with no additional training. For the supervised warm start on base-model successes (S32), the pooled step-0 gain against the same-budget Dr. GRPO trajectory is +4.77 pass@16 points at 512 tokens, 90% $[+3.45, +6.07]$, and +2.15 at 1,024 tokens, 90% $[+1.10, +3.20]$. Under the reading fixed before re-evaluation, a gain below half its original magnitude is classified as largely a budget effect. The completed check meets that condition while retaining a smaller positive gain. Later S32 checkpoints lie outside the available reference accuracy range at both budgets, so this comparison concerns the supervised initialization.

The registered Check 21 comparison uses checkpoints with common reference support at both budgets and paired bootstrap draws. The interim seed 10 observation reverses the ordering: the human-solution arm is below the same-budget Dr. GRPO trajectory at 1,024 tokens. [Check 21: remaining seeds and the registered pooled budget-effect verdict.] Separately, Preregistration 19 rule 1 is not interpretable because HQ checkpoints in seeds 9 and 10 do not overlap the HP accuracy range. The descriptive step-150 zero-count stratum contrast, HQ minus HP, is +0.58 pass@16 points, 90% $[-2.13, +3.43]$, and cannot substitute for the registered matched-accuracy test.

Matched pass@1 controls mean success, but does not control which answers are censored by a length limit. Claims about an accuracy–diversity frontier must state the answer budget, use supported comparisons within that budget, and assess sensitivity to differential truncation. Re-evaluation changes the measurement protocol; Section 8 reports replication with training and evaluation at 1,024 tokens.

8. Replication at a larger budget and a larger model

Pre-registration 21 trains and evaluates Qwen2.5-Math-1.5B at 1,024 tokens in seeds 8–10. Each arm uses 150 steps and 128 rollouts per step, with problem order paired to its 512-token counterpart. Both replications change the answer cap and loss normalizer together, so neither isolates the causal effect of the cap. With B1k denoting Dr. GRPO, matched-pass@1 differences in pass@16 are NSR1k–PSR1k -0.25, 95% $[-0.72, +0.20]$; PSR1k–B1k $+0.11$, 95% $[-0.26, +0.49]$; NSR1k–B1k -0.15, 95% $[-0.62, +0.31]$; and AC1k against the B1k envelope -0.12, 95% $[-0.60, +0.29]$. All four meet average equivalence, supporting the registered joint statement that the core result replicates at 1,024 tokens. Equivalence requires pooled and per-seed 90% intervals within ± 2 points; bootstrap intervals resample problems conditional on the trained seeds.

Step-150 truncation is 15.2–17.0%, below the 20% reporting qualifier. Comparison points remain steps 50, 100, and 150; step 25 supplies reference knots only. Omitting these knots gives PSR1k–B1k $+0.14$, 95% $[-0.24, +0.53]$; NSR1k–B1k $+0.44$, 95% $[-0.07, +0.97]$; and NSR1k–PSR1k $+0.33$, 95% $[-0.22, +0.91]$. The early NSR comparison therefore depends on reference resolution.

At 7B and 2,048 tokens, Pre-registration 20 uses seeds 8 and 9. PSR7–B7 is -0.09, 90% $[-0.48, +0.31]$, and NSR7–B7 is -0.09, 90% $[-0.45, +0.27]$; both reach average equivalence. AC7 against the B7 envelope gives -0.15, 90% $[-0.57, +0.16]$, 95% $[-0.64, +0.23]$, replicating the allocation null. NSR7–PSR7 gives $+0.16$, 90% $[-0.29, +0.57]$, but 13.7% of bootstrap points in seed 9 are clamped, exceeding the 10% guard. This contrast is not interpreted: three of four contrasts replicate, leaving the full joint claim unsupported. Both replications retain frozen AC settings; their null results concern that allocation rule.

Late test-set pass@16 is compressed near 94–95, limiting sensitivity. The separately reported hard cohort contains 253 problems selected by independent base screening for zero to two successes in eight samples, with labels audited before evaluation. Screening samples are not reused for evaluation, and label disagreements are removed without replacement. The trained B7 checkpoints there span pass@16 80.8–84.5. Only NSR7–B7 reaches average equivalence; PSR7–B7 is inconclusive, and the other two contrasts fail the clamping guard. The frozen frontier model meets rule 3 in aggregate, with RMSE 0.23 over eight B7 seed-checkpoints. Its observed-minus-predicted pass@16 errors offset: $+7.0$ to $+22.9$ points for zero successes in 32 base samples and -6.8 to -2.7 for one to four.

The evaluations use vLLM following registered amendments. The equivalence check gave base-7B pass@1/pass@16 of 30.5/88.0 and 30.1/86.8 in independent runs, versus 29.9/87.5 with Hugging Face. For B1k seed 9, step 50 gave 52.2/87.5 and 51.9/86.4 versus 52.5/86.7; step 150 gave 58.1/86.7 twice versus 58.0/86.8. Truncation and lengths also agreed. Every replication comparison uses one engine, including re-evaluated base models and references; the frozen rule 3 forecast retains its original Hugging Face base counts.

9. A per-problem model of the frontier

We test a descriptive model of the frontier using empirical-Bayes estimates from 32 base-model samples to initialize each problem's log-odds. With $p_x = \sigma(\theta_x)$, the dynamics are

$$\dot{\theta}_x = p_x(1 - p_x) + \tau \, \tfrac{1}{N} \textstyle\sum_{x'} p_{x'}(1 - p_{x'}),$$

combining state-dependent own-problem growth with a uniform transfer term. We fitted $\tau = -0.20$ on Dr. GRPO's nine checkpoints and froze it before computing the registered test predictions. Each checkpoint's observed pass@1 determines its location on the model trajectory; pass@16 and stratum-level success are then predictions. The model therefore describes the frontier conditional on accuracy, rather than predicting training speed.

The frozen model has pass@16 RMSE of 1.21 points for nine further Qwen rules (56 available checkpoints), 1.90 points for five OLMo rules with τ transferred from Qwen (50 checkpoints), and 0.85 points for four 1,024-token evaluation checkpoints (Figure 2). The affine log-odds baseline has RMSE of 2.45, 12.58, and 1.27 points on these sets, respectively; the uniform-shift baseline ranges from 9.85 to 20.07 points. OLMo's aggregate RMSE coexists with larger mean stratum residuals: observed minus predicted pass@16 is $+6.0$ points in the zero-count stratum and -6.8 in the rare stratum. On Qwen, a uniform shift predicts pass@16 of

70–80 for problems with zero successes in 32 base samples, compared with an observed 24.0 for Dr. GRPO at step 150.

The model fails the registered warm-start test when initialized from the warm-start policy's own step-0 counts: its pass@16 RMSE is 3.56 points. The longer-training test remains incomplete, with the available Dr. GRPO reference measured through step 250 and the registered extrapolation verdict pending. Deep base sampling is also partial: three of four shards cover 232 of the 309 problems with at most four successes in the initial 32 samples. Using 256 additional samples per problem, the refitted coefficient is $\tau = +0.20$, with a 90% interval $[+0.10, +0.35]$ from 200 bootstrap draws. Refining the initial success estimates changes the fitted coefficient from negative to positive; its interpretation as transfer therefore depends on the initialization and the assumed log-odds dynamics.

Pre-registration 18 tests the model's response to a deliberate change in rollout allocation. In the registered point forecast comparison, both model variants confirm the HR and AV forecasts, but neither confirms AC: its observed mean matched-accuracy gain is -0.05 points, against $+4.12$ from the frozen law and $+2.46$ from deeper initial counts. The allocation diagnostics establish that AC supplies additional successful samples to pilot-zero groups, so the failed forecast accompanies a realized change in the sampling mechanism. Deeper initialization counts reduce the forecast discrepancy without resolving it. The model's agreement with aggregate pass@16 in the earlier audits therefore supports a conditional descriptive fit, while its use to guide this allocation intervention is unsupported. The warm-start failure and sensitivity to initialization further limit a mechanistic interpretation. The expected-update identities constrain gradient estimators; they do not validate the assumed dynamics of learning and transfer across problems.

10. Process credit from minimum hints

Pre-registration 23 tests process credit on the policy's own failed prefixes. During the first 100 training steps, it probes up to four failed answers on each of up to four problems with at most one success in eight rollouts. Prefixes contain at most 256 tokens and end before the answer box. Each receives zero, 32, or 64 tokens from a verified human solution, cut before its answer, with two continuations per hint level under a total 1,024-token budget. Minimum-help credit (MH) assigns values 1, 2/3, or 1/3 for the smallest tested hint level yielding a verified success, and 0 if none succeeds. Values are centred within each problem, and credit

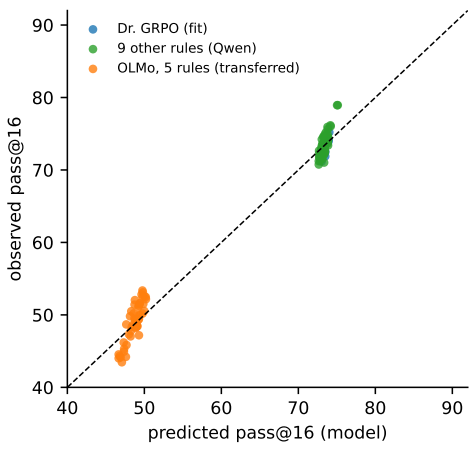


Figure 2. Predicted against observed pass@16 for the frozen per-problem model.

updates only policy-generated prefix tokens. Its coefficient $c_t = 2(1 - t/100)$ decreases linearly to zero; Dr. GRPO training continues to step 150. The averaged-value control (AVH) uses identical hints, queries, and schedule but scores each prefix by mean success across its six continuations.

On 255 validation problems in seed 8, MH against the B1k envelope gives -0.01 pass@16 points, 95% $[-1.24, +0.89]$; AVH gives -0.05, 95% $[-1.32, +0.91]$. MH against the AVH trajectory gives $+0.05$, 95% $[-0.67, +0.84]$. Both mechanism gates are met: MH credits 520 prefixes, exceeding the 150 floor, and values differ in 61% of probed problem-steps, above the 30% gate. AVH credits 563 prefixes with contrasts in 64%. Neither reaches the registered $+1.5$-point gain gate, so selection drops the method without a three-seed test. These intervals remain conditional on one training seed.

11. Discussion

The tested correctness-only interventions mainly change how quickly training reaches a given accuracy. At matched pass@1, the registered comparisons support a shared pass@16 curve within each setting. The larger-budget replication reaches average equivalence in all four contrasts; the 7B replication does so in three, while its direct NSR–PSR comparison is excluded by the clamping guard. This is agreement within the registered tolerance over supported checkpoints. It permits smaller differences and does not establish equivalence at every checkpoint. The small replay gain also remains below the registered frontier-improvement threshold.

Similar pass@16 at matched pass@1 means similar repeated-sampling coverage at a fixed mean success probability. It does not identify the full distribution of per-problem success or the properties of generated solutions. Nor does it establish invariance for other task families, longer training, or non-correctness rewards. Minimum-help credit supplies external process information, but its failed selection in one validation seed cannot settle that broader question. Answer budgets remain part of the estimand: differential truncation can create apparent gains, while pass@16 compression at 7B reduces sensitivity. [Check 21: pooled human-solution budget-effect verdict.] [P22: external versus self prefix guidance.]

The update identities do not imply the observed invariance. Per-problem coefficients can rotate the aggregate gradient through shared parameters, and optimizer state and noise can alter the realized update. Nor does aggregate prediction identify a learning mechanism: the 7B model fit combines small aggregate error with large offsetting stratum errors.

The forthcoming causal update-direction test is therefore the natural next step ([P25]). Perturbing direction at a common policy while controlling update magnitude and answer budget would ask whether updates follow the same local pass@1–pass@16 relation. Measuring both quantities after the intervention would connect direction to frontier displacement directly. This would test whether the shared curve reflects a common effective update direction, beyond the descriptive agreement of completed training trajectories.

12. Limitations

Our experiments use 1B and 1.5B models, with main runs of 150 steps on Qwen and 300 on OLMo. Generalization to larger models and longer training horizons remains untested. The evaluation budget matters substantially: for Dr. GRPO seed 10 at step 150, increasing the budget from 512 to 1,024 tokens raises pass@1 from 44.6 to 57.7 and reduces truncation from 48% to 15.5%. The 1,024-token check remains partial. Bootstrap intervals cover problem sampling conditional on the observed training seeds; they do not estimate training-run variability. Average equivalence within ± 2 points permits smaller and checkpoint-specific differences. The longer-reference comparisons in Section 5 are interim estimates against one seed and include only overlapping checkpoints.

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A. Proofs

Theorem 3.1. Couple the rule to an i.i.d. sequence $y_1, y_2, \ldots$ with rewards r. Given r, the stopping time N and the advantages A_i are fixed, the answers are independent, $\mathbb{E}[\ell(y_i) \mid r] = \kappa(r_i)\nabla p$ with $\kappa(1) = 1/p$ and $\kappa(0) = -1/(1-p)$, and $\mathbb{E}[\|\ell(y_i)\|^2 \mid r] = F_{r_i}$. Hence $\mathbb{E}[g \mid r] = \sum_{i \le N} A_i \kappa(r_i)\nabla p$ and $\mathbb{E}[\|g\|^2 \mid r] = \sum_i A_i^2 F_{r_i} + \sum_{i \ne j} A_i A_j \kappa(r_i)\kappa(r_j)\|\nabla p\|^2$; averaging over r, whose law depends only on p, gives the claim. Proposition 2.1. $f_k''(p) = -k(k-1)(1-p)^{k-2} < 0$ for $k > 1$; apply Jensen's inequality. Lemma 3.3. With all rewards equal, $r_i - \bar{r} = 0$ for every i; a reweighting that depends only on the rewards multiplies these zeros.

B. Pre-registrations

Registration	Comparison	Outcome
9	OLMo NSR–PSR replication	Average matched-accuracy equivalence; larger NSR second moments in both seeds. The registered speed prediction failed because one seed tied.
10	Qwen noise injection and learning-rate controls	Pooled early-lag reproduction with seed heterogeneity; average matched-accuracy equivalence. NSR still lagged at the matched doubled learning rate; gradient-norm persistence was inconclusive.
11	Late-checkpoint temperature control	Mean matched-pass@1 difference -1.42 points, 90% $[-2.19, -0.68]$.
12	Coverage-weighted replay against Dr. GRPO's envelope	$+1.19$ points, 95% $[+0.54, +1.79]$: average agreement. Registered frontier improvement, equal-cost accuracy improvement, and a coverage-weighting benefit were not established.
14	Base-seeded replay and controls	Interim base-seeded estimate $+1.02$ points, 95% $[+0.23, +1.68]$. Some registered contrasts remain incomplete; larger-group and warm-start mean endpoints exceed the original reference range.
17	Frozen frontier model	Available Qwen-method, OLMo, and 1,024-token test sets meet the aggregate RMSE criterion. The warm-start test fails; longer-training extrapolation remains pending.
16 (check)	Answer budget: re-evaluation at 1,024 tokens	Warm-start step-0 gain $+4.77$ at 512 and $+2.15$ at 1,024 tokens: largely a budget effect.
18	Rollout allocation at a fixed budget (AC, AV, HORA)	Rules 1–3 not met; AC $+0.14$, 95% $[-0.36, +0.58]$; forecasts confirmed for HR and AV, not for AC.
19	Supervised fine-tuning on human solutions	Rule 1 not interpretable (insufficient overlap); zero-count stratum contrast $+0.58$, 90% $[-2.13, +3.43]$.
20	7B replication (2,048 tokens)	PSR7–B7 and NSR7–B7 reach average equivalence; AC7 replicates the allocation null (-0.15, 95% $[-0.64, +0.23]$). NSR7–PSR7 is not interpreted: 13.7% clamping exceeds the 10% guard. Three of four contrasts replicate. Rule 3 RMSE is 0.23, with large offsetting stratum errors.
21	1.5B replication at 1,024 tokens	All four contrasts reach average equivalence: the core result replicates. AC1k against B1k's envelope gives -0.12, 95% $[-0.60, +0.29]$. Truncation is 15.2–17.0%; step-25 reference knots affect the early NSR comparison.
21 (check)	Human-solution arms at 1,024 tokens	[pending]
22	External versus self prefix guidance	[pending]
23	Minimum-help credit (pilot)	Against B1k's envelope, MH gives -0.01, 95% $[-1.24, +0.89]$, and AVH -0.05, 95% $[-1.32, +0.91]$. MH against AVH's trajectory gives $+0.05$, 95% $[-0.67, +0.84]$. Mechanism gates met: 520 MH prefixes, contrasts in 61%. Neither arm reaches the $+1.5$ gain gate; no candidate, method dropped.

Pre-registration 12's nominal second-half confirmation set cannot provide independent confirmation: those 256 problems had already been used for branch selection in Pre-registration 11. Interim results from Pre-registration 12's seeds 9 and 10 were visible when Pre-registration 14 was declared. In the Pre-registration 12 bootstrap, endpoint clamping occurred for 0 of 18,000 coverage-weighted replay points, 35 of 18,000 uniform-replay points, and 267 of 18,000 unlikeliness points. These intervals cover problem sampling conditional on the observed training

seeds.

C. Label audit

The 511-problem Qwen evaluation set used audited answer keys. In the OLMo analysis, excluding one audit-rejected problem did not change any registered verdict.